

CARLOS MORENO ROJAS

Biomedical & Electrical Engineering · Engineering / R&D; CV

Engineer focused on the design, integration and validation of technical systems combining electronics, embedded systems, sensors, software, biomedical instrumentation, FPGA, fluidics, CAD and prototyping. Experience ranges from medical devices and robotics to experimental research platforms.

PROFESSIONAL EXPERIENCE

Candel Medical Company

Professional internship

Hardware, electronics and firmware for transcranial electrical stimulation; circuit analysis, PCB work, embedded systems, BLE, signal generation and validation.

Meldic Ltda.

Medical equipment experience and technical development projects.

SELECTED ENGINEERING PROJECTS

CYBATHLON 2024 - Ciervo UC

Electrical Area Leader

Development and integration of electronics for a robotic transfemoral prosthesis.

Ex Vivo Liver Perfusion Platform

Master's Research

Fluidic circuit, sensors, physiological monitoring, acquisition, calibration and system validation.

Universal Testing Machine Upgrade

FPGA / Zybo Z10

Vivado and VHDL implementation for force, displacement, deformation and speed variables.

Organ Preservation Machine

Mechanical, fluidic and control-system redesign.

Vscan Air Enclosure

CAD, design iteration and physical prototyping.

Lung Segmentation - U-Net

Deep-learning medical image segmentation.

CORE ENGINEERING SKILLS

- Analog/digital electronics, PCB and sensor integration
- ESP32, STM32, Arduino, ADC/DAC and communications
- Python and C++
- FPGA: Zybo Z10, Vivado and VHDL
- Pressure, flow, temperature, PT100/PT1000 and acquisition systems
- CAD, Fusion 360, 3D printing, machining and electromechanical integration
- Experimental calibration, testing, troubleshooting and validation

EDUCATION

Electrical Engineering

Pontificia Universidad Católica de Chile · 2025–2026

MSc Biomedical Engineering

Pontificia Universidad Católica de Chile · 2025–2027

Biomedical Engineering

Pontificia Universidad Católica de Chile · 2021–2024

LEADERSHIP & TEACHING

Biomedical Engineering Student Chapter UC

President · 2024–2025

IEEE EMBS PUC Chile

Webmaster · since 2024

Physics Laboratory Teaching Assistant

FIS154

LANGUAGES

- Spanish - Native
- English - B2 / ALTE 3